Vishal Dole

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EDUCATION

Indian Institute Of Information Technology, Allahabad **2022 - 2024**
M.Tech - Information Technology. ***CGPA : 8.52*** *Prayagraj, India*

Kavikulguru Institute of Technology and Science, Ramtek **2016 - 2020**
B.E - Information Technology. ***CGPA : 8.15*** *Ramtek, India*

EXPERIENCE

Tata Consultancy Services (TCS) **Jan 2025 – Present**
Software Engineer *Bangalore, India*

Client: Central Bank of Kuwait
Project: Banking Operations Copilot for Internal Employees

- Developed an **Account Opening Verification Agent** using **Agentic RAG**, **LangChain**, vector search, and LLM-based document validation to verify onboarding documents against banking SOPs, detect compliance gaps, and automate operational decision support.
- Built a **KYC Renewal Agent** using the **Model Context Protocol (MCP)** for tool orchestration, enabling AI-driven customer profile analysis, compliance validation, missing document detection, and intelligent KYC renewal recommendations.
- Engineered a **Loan Processing Agent** integrating LLM reasoning with policy-based decision workflows to evaluate borrower eligibility using income verification, credit risk assessment, debt-to-income analysis, and collateral validation for underwriting automation.
- Designed a production-ready **multi-agent Banking Operations Copilot** using **Agentic AI**, and enterprise **RAG**, orchestrating specialized AI agents with workflow automation, context-aware reasoning, and grounded responses from internal banking knowledge repositories.

PROJECTS

Agent-Based RAG Chatbot for Insurance Policy Analysis

- Developed an Agent-Based RAG chatbot for insurance policy analysis, enabling semantic retrieval across **500+** pages of policy documents with grounded responses.
- Implemented Adaptive RAG and Corrective RAG workflows to rewrite queries, validate retrieved context, and eliminate **30%** of low-relevance document chunks.
- Built a multi-agent workflow with Retriever, Web Search, Hallucination Validator, and Response Generator agents using LangChain, LangGraph for reliable question answering.
- Enhanced response quality through semantic search, reranking, and hallucination detection, reducing unsupported responses by **40%** while incorporating real-time web information.

Warden - Intelligent Compliance Platform

- Engineered an **Agentic RAG** application using **FastAPI**, **LangGraph**, and **LLM orchestration** to automate multi-step compliance evaluation, reasoning, and report generation.
- Developed a **Next.js** and **TypeScript** dashboard integrated with FastAPI microservices for real-time workflow monitoring, AI-generated insights, and compliance reporting.
- Implemented **stateful AI workflows** with caching, retry mechanisms, asynchronous task execution, and fault-tolerant orchestration to improve scalability, reliability, and inference efficiency.
- Designed and deployed secure **RESTful APIs** supporting **10+ compliance rule domains**, incorporating observability, structured logging, authentication, and production-ready enterprise deployment practices.

TECHNICAL SKILLS

Languages	C, C++, Python, JavaScript, TypeScript, Node.js
Framework	Django, Flask, Pytest, React, Express.js
Database	MySQL, Pinecone, MongoDB
Cloud & DevOps	Git, GCP, Docker, CI/CD (GitHub Actions, Azure Pipelines)
AI/ML	Generative AI, RAG pipelines, LLM integration, LangChain, LangGraph, LoRA, QLoRA

ACHIEVEMENTS

- Solved **500+** problems on **LeetCode**, **GeeksForGeeks** and other platforms.